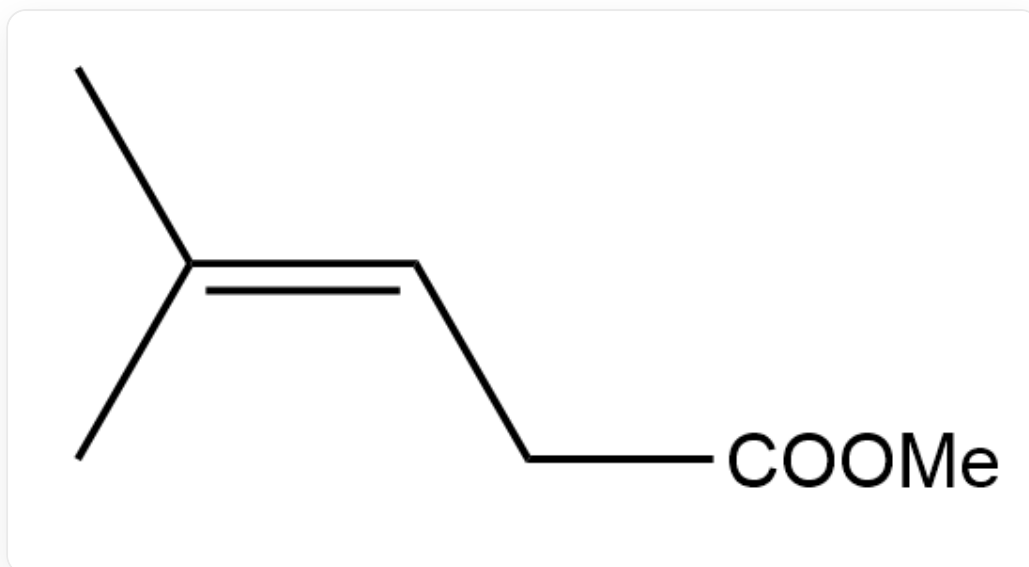


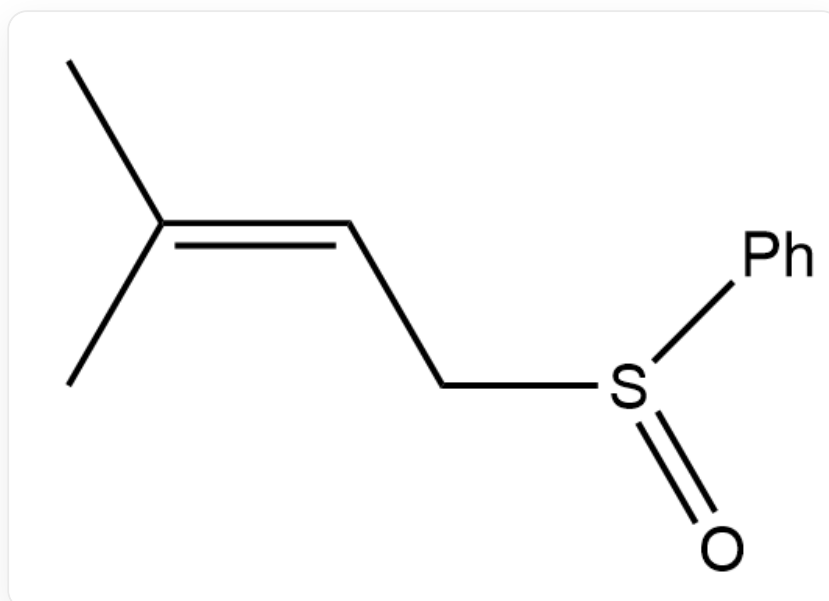
题目

单线态氧¹O₂可以与以下两种底物发生反应，请分别给出该加成反应的主产物（不考虑立体异构）



C/C(C)=C/CC(OC)=O,底物1

底物1

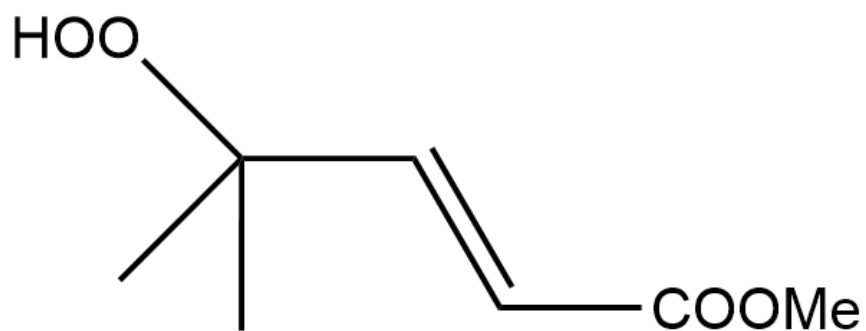


C/C(C)=C/CS(C1=CC=CC=C1)=O,底物2

底物2

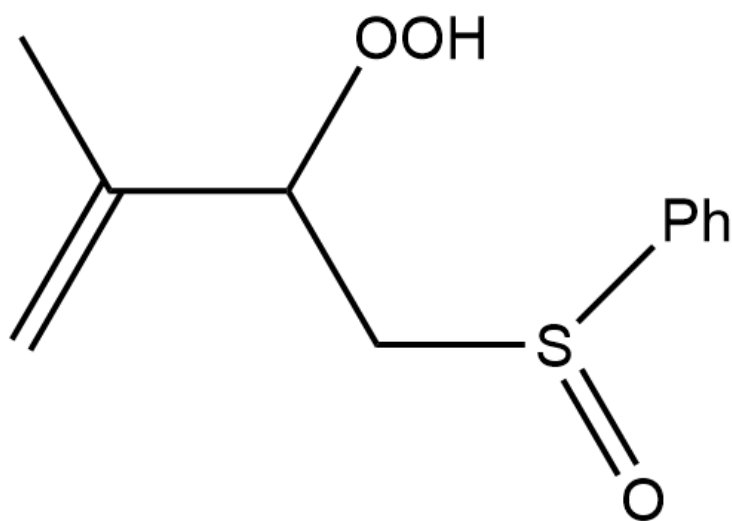
A. 其他选项均不正确

B.



CC(C)(OO)/C=C/C(OC)=O,产物1

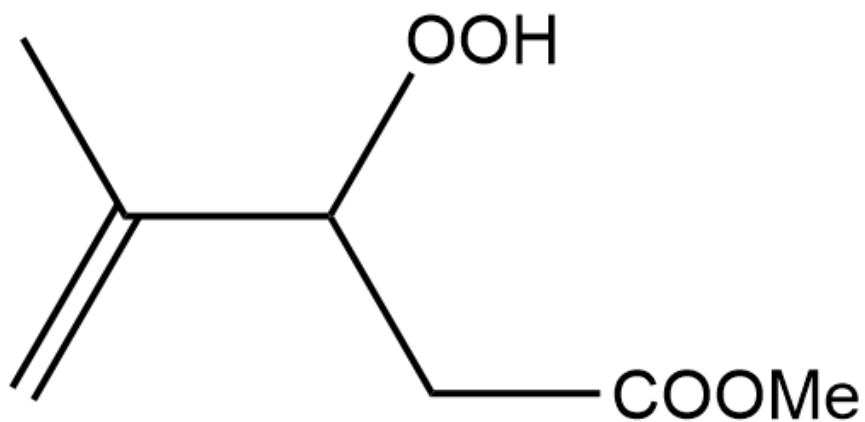
产物1



CC(C(OO)CS(C1=CC=CC=C1)=O)=C,产物2

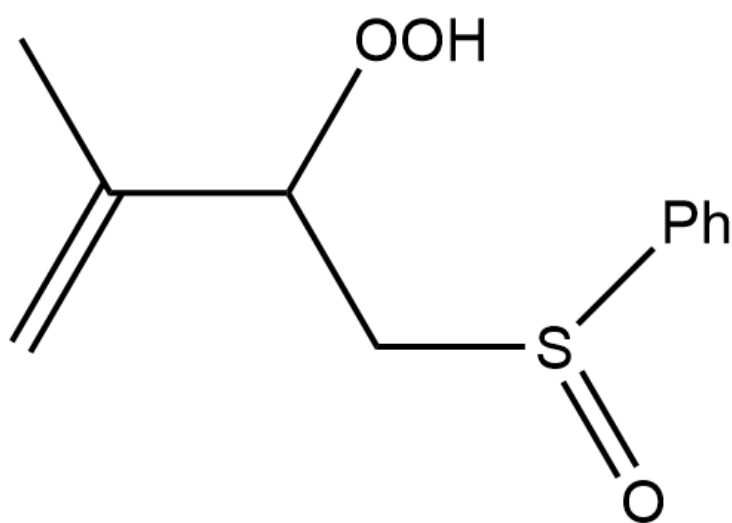
产物2

C.



CC(C(OO)CC(OC)=O)=C,产物1

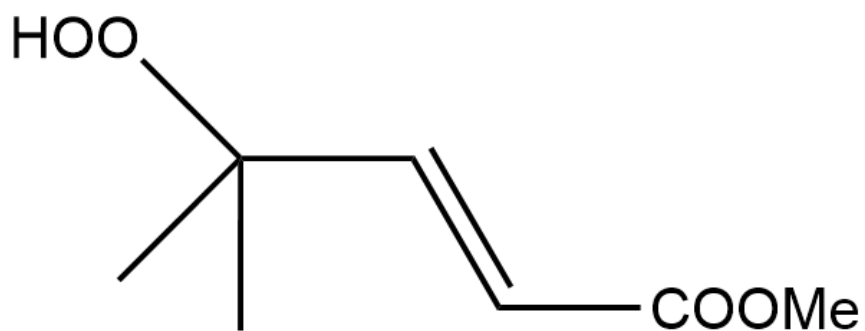
产物1



CC(C(OO)CS(C1=CC=CC=C1)=O)=C,产物2

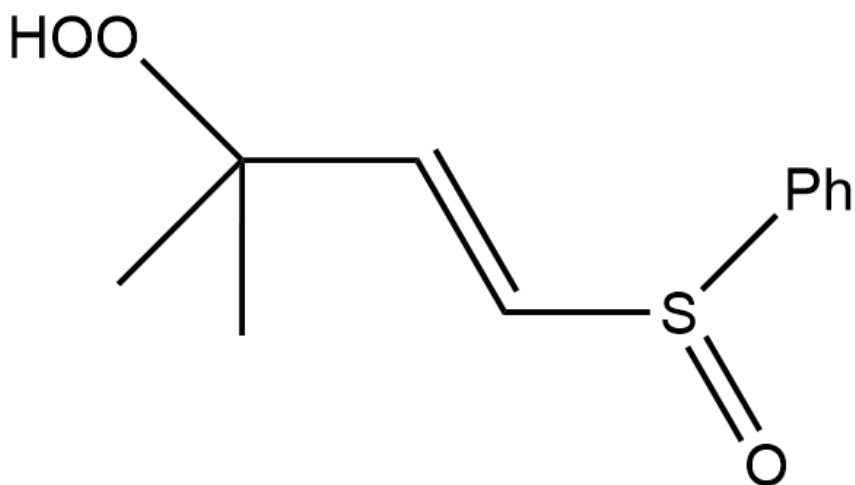
产物2

D.



CC(C)(OO)/C=C/C(OC)=O,产物1

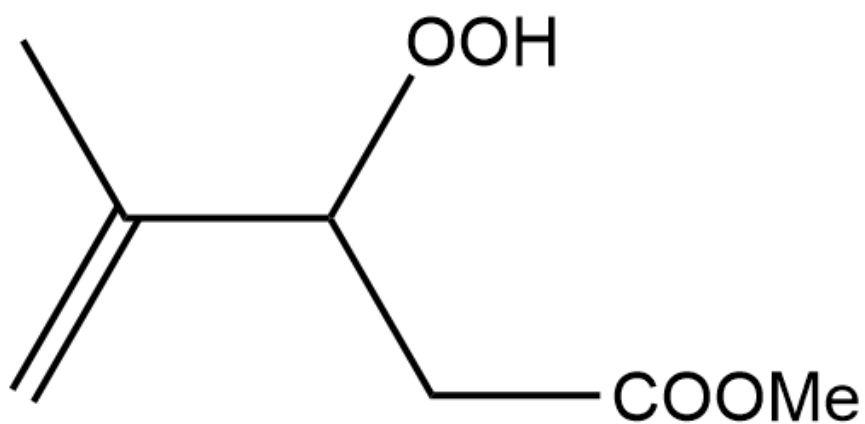
产物1



O=S(C1=CC=CC=C1)/C=C/C(C)(OO)C,产物2

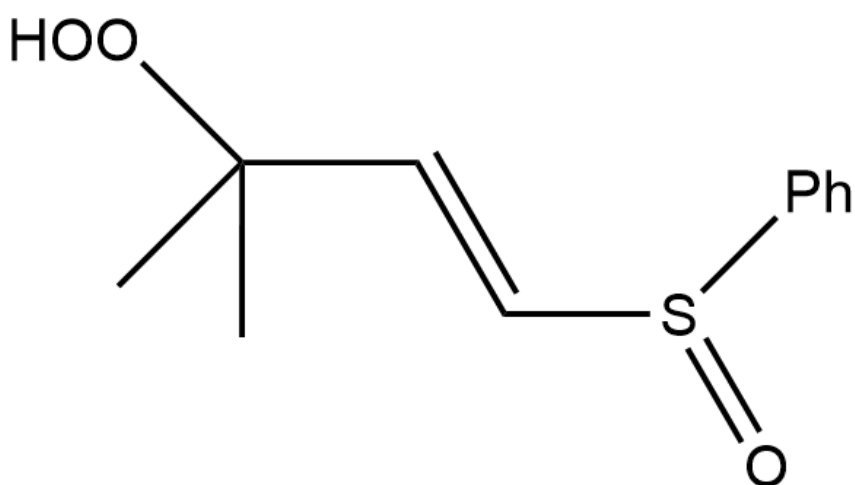
产物2

E.



CC(C(OO)CC(OC)=O)=C,产物1

产物1



O=S(C1=CC=CC=C1)/C=C/C(C)(OO)C,产物2

产物2

答案

正确答案: **B**

详细解析

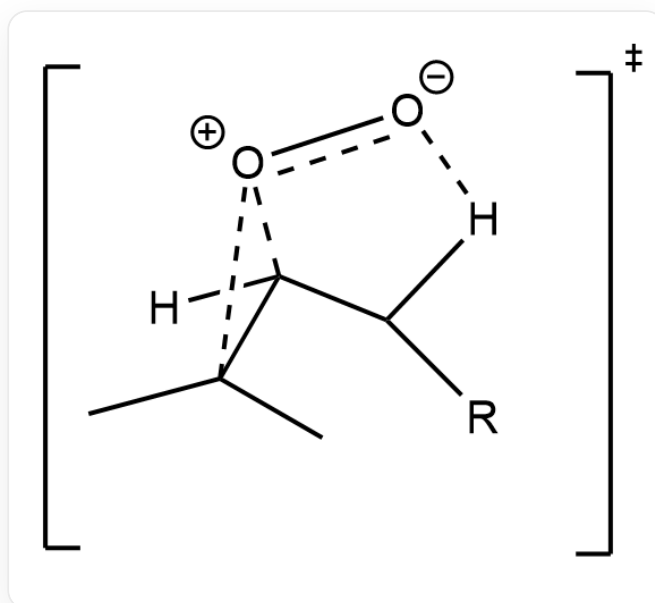
亚砷与羰基相比其极化程度更大,氧上负电荷更集中

CHECKPOINT

1 PTS

亚砷与羰基相比其极化程度更大,氧上负电荷更集中

这是反应的过渡态（以反应1为例）



[H]C1(C([O+])1[O-])(C)C([R])[H]

对于R=COOMe,进攻羰基邻位的氢,羰基可通过共轭效应稳定过渡态,故得到共轭产物

CHECKPOINT

1 PTS

对于R=COOMe,进攻羰基邻位的氢,羰基可通过共轭效应稳定过渡态,故得到共轭产物

对于R=SOPh,氧上的负电荷过多,使得进攻其邻位氢的过渡态中过氧基和亚砷氧间存在强烈排斥,故倾向生成末端烯烃

CHECKPOINT

1 PTS

对于R=SOPh,氧上的负电荷过多,使得进攻其邻位氢的过渡态中过氧基和亚砷氧间存在强烈排斥,故倾向生成末端烯烃